MUST CS460 Computer Vision Final Project Guideline

Choose a question about image classification, recognition, detection, or segmentation. The datasets and questions in this guide are examples only. A pattern you notice in the images can be a good reason to investigate a different question.

What the project requires

Develop your own research question. Explain what a model receives, what it predicts, and what you want to learn. An unexpected result or an assumption you want to challenge can guide your next experiment.

Personally train or fine-tune a deep learning baseline and evaluate at least one meaningful variant. Training only a classifier head on frozen features counts. A variant changes something concrete about the starting method so you can test an idea. You may explore as many variables as your question warrants, separately or together.

Explain what the evidence supports. When several settings change together, describe their combined effect. A thoughtful investigation can earn full credit even if your variant performs worse.

What could you change?

Training images. Start with a CNN trained on unmodified aerial images. Train a variant that sees random rotations. Does it recognize the same scene when the test image is turned?

Layers that learn. Start with a pretrained ResNet-18 and train only its classifier. In a variant, also update the last residual stage. Does this help distinguish similar flower species?

Training loss. Start with ordinary cross-entropy on an imbalanced dataset. Give rare classes more weight in the variant's loss. Check which rare examples it recovers and which new errors it makes.

Input resolution. For a detector, change the input size while keeping the boxes aligned. Does the smaller input save time at the cost of missing distant signs? **All example topics, methods, and layouts in both documents are for reference only.**

Computing resources

You can complete the project without paying for a GPU. A CPU and a fixed subset of NEU-CLS or Fashion-MNIST are suitable starting points. Time one epoch before committing to a run. If needed, reduce the image size, freeze a backbone, or use a smaller training subset.

Free GPU notebooks are available through Kaggle and Google Colab. Check your available quota and save checkpoints. For optional paid access, AutoDL offers metered instances. Check the rate before starting and shut down the instance after exporting your results. Expanded storage may be billed separately.

Submission checklist

Submit the following items together. The sample images make the project quick to test. Report final metrics on your stated evaluation split and identify that split in the paper. **Please note that the uploaded file size must not exceed 50 MB**. If it does, please provide the data samples or checkpoints via a cloud drive link.

- **English report (PDF).** State your question, baseline, variant, findings, and interpretation. Include a training-loss curve from your own run. Choose the length, metrics, and presentation that fit your question.
- **Runnable code.** Include training, evaluation, and sample-prediction scripts or a notebook, plus the configuration used for the reported runs. Include the checkpoint needed for the sample run, or give an accessible download link if it is large.
- **Small test-data pack.** Supply a few held-out inputs with their ground truth (class labels, bounding boxes, or masks). Five to ten images are usually enough for a quick check. Keep the pack small. If the dataset restricts redistribution, provide the sample IDs and a script or instructions that retrieve those exact items.
- **Test-set examples with predictions.** Reuse images from the small pack. Save the model output alongside the sample ID and ground truth, for example as a CSV of predicted classes or an annotated detection image. Identify the run or checkpoint that produced it.
- **Training evidence.** Include the original machine-generated loss log for the run shown in the report. A CSV, TensorBoard event file, or notebook log is suitable. Use the same run name in the log, configuration, and figure caption.
- **README.md.** Give dependency versions, data and weight locations, and exact commands or notebook cells for training, regenerating the curve, and testing the small pack. State the expected output for a named sample. Credit borrowed code and briefly identify any AI assistance, what you implemented or modified, and how you verified it.

How your training will be checked

Plot the loss your training code recorded against epoch or optimizer step. Name the run and the parameters you trained. In your own words, identify a point or interval on the curve and explain one training choice in light of what you saw. A noisy or flat curve is acceptable.

Write the report in your own words. You should be able to explain the code that updates the weights. **Be ready for a short spot check.** You may be asked to regenerate the curve from the log or run the saved model on a supplied test image and explain its prediction.

A worked example in defect recognition

Illustrative example. A student notices that fine scratches are harder to see in thumbnail images of steel. They ask whether reducing image resolution also makes a CNN miss more scratches. The choices below show how that particular observation could become an experiment.

Question and hypothesis

The input is a grayscale image of steel. The output is one of six defect classes. The student expects thin marks to fade after downsampling. Overall accuracy summarizes classification across all defects, while scratch recall directly addresses the question about missed scratches.

One possible experiment

Step	Illustrative choices
Prepare the data	NEU-CLS has 300 images per class. The student uses a fixed seed to assign 210 per class to training, 45 to validation, and 45 to testing. Both runs use the same 270 test images.
Build a variant	The baseline takes 128 × 128 inputs. The variant takes 64 × 64 inputs. Both use the same three-layer CNN, adaptive average pooling, initial weights, and training settings so a difference is easier to interpret.
Select checkpoints	The student logs training loss and validation accuracy, then selects each run's checkpoint using validation accuracy before evaluating on the test split.
Count predictions	The student calculates overall accuracy using all 270 images and scratch recall using the 45 true scratch images. These measures address different aspects of this question.
Inspect errors	The student compares a scratch the baseline recognized and the variant missed, if available. Looking for cases where the variant did better helps test competing explanations.

How to interpret a difference

Suppose the baseline identifies 40 of 45 scratches and the variant identifies 36. In this **hypothetical result**, recall falls from 88.9% to 80.0%, a drop of 8.9 percentage points. Comparing image IDs reveals which scratches the variant missed and whether it also recovered some that the baseline missed.

Where your own investigation might lead

Suppose many missed scratches also have low contrast. You might then vary both resolution and contrast augmentation to see whether the latter helps at smaller sizes. Another student might examine inference time instead. **Let what you observe guide what you try next.**

Datasets for classification and recognition

You may choose a public dataset outside this guide. The linked classification and recognition datasets illustrate different scales and label types. Look at a few examples from any dataset you select, check that its labels fit your question, and cite its source.

Dataset	Images and labels	Possible research question
NEU-CLS	1,800 grayscale steel images across six defect classes, with 300 per class.	Do fine scratches and broad patches respond differently to lower input resolution?
Fashion-MNIST	Ten clothing classes in 28 × 28 grayscale images. The official split has 60,000 training and 10,000 test images.	When the training set grows from 100 to 500 images per class, does confusion between shirts and T-shirts decrease?
Beans	1,295 field photographs of healthy leaves, angular leaf spot, or bean rust.	Which disease class loses the most recall when RGB is replaced by grayscale? Repeat the gray channel three times to preserve the model input shape.
TrashNet	2,527 waste images in six classes. Class sizes vary and the backgrounds are fairly uniform.	Does a class-weighted loss improve recall for the small trash class? Check whether other materials are now mistaken for trash.
UC Merced Land Use	2,100 aerial images across 21 land-use categories, with 100 per class.	Does random 90-degree rotation during training help recognize rotated scenes? Also evaluate the original test images.
Oxford Flowers 102	102 flower classes. The official training and validation splits each contain ten images per class.	With a pretrained ResNet-18, how does training the classifier alone compare with also fine-tuning the final residual stage?

Optional extensions

Compare a frozen ResNet-18 with a small frozen vision transformer. Cache image features and fit a linear classifier to each. If the backbone and pretraining data both differ, your conclusion applies to those two pretrained models. One available model family is DINOv3.

For a GAN or diffusion project, test whether synthetic training images improve a classifier on real held-out images. Compare ordinary augmentation with ordinary augmentation plus synthetic images, keeping the classifier and number of training steps fixed.

Datasets for detection and segmentation

For detection, the model predicts a class and bounding box for each object. For segmentation, it predicts a label for each pixel. **The questions below are examples. You may also find and use your own public dataset.** Inspecting annotations on the original images can help you identify a different question worth pursuing.

Object detection

Dataset	Images and labels	Possible research question
Penn-Fudan	170 street images with 345 pedestrian instances. Instance masks can be converted to boxes.	In a pretrained Faster R-CNN, does unfreezing the final backbone stage improve pedestrian AP50?
GTSDB	900 road images in a 600-image training split and a 300-image evaluation split, with sign locations.	Merge annotated sign types into one class. At a fixed score threshold, how often does a smaller input miss small signs? Define "small" in original image coordinates.
BCCD	364 microscopy images with boxes for red cells, white cells, and platelets.	Does reducing input resolution affect platelet recall more than red-cell recall? Use the same matching rule in both runs.

Image segmentation

Dataset	Images and labels	Possible research question
Oxford-IIIT Pet	About 7,400 pet images with a trimap marking foreground, background, and an ambiguous boundary.	Does higher resolution help a small U-Net recover thin structures? For binary masks, ignore the ambiguous band in both training and evaluation.
Kvasir-SEG	1,000 endoscopy images with a binary polyp mask for each image.	Compare binary cross-entropy and Dice loss in the same U-Net. Does Dice change for polyps that occupy a small fraction of the image?
SUIM	Eight underwater pixel classes. There are 1,525 training/validation images and 110 test images.	Treat divers as foreground and all other classes as background. Does color augmentation improve foreground IoU?

Data splits and subsets

Keep the official held-out split when available. Otherwise, document your split rule and save the image IDs. Group related frames, patients, or source scenes when identifiers are available. Split original images before augmentation. If you use a subset, record how you sampled it and the number of images retained in each split.

How to write the report

Use a structure and length that suit your project. The headings in the report template suggest one way to organize the paper. Give readers enough detail to follow your decisions, understand the evidence, and judge your conclusions.

Title and abstract

Give the paper a title specific to your investigation. In the abstract, summarize the question, approach, main finding, and its significance. Drafting it after the experiments makes it easier to report what actually happened.

1 Introduction and Research Question

Describe the task, its input and output, and the question you chose. **Explain what prompted the question** and what you expected to find. Briefly credit prior work that influenced your starting point.

2 Data and Annotations

Describe the dataset, labels, subset, and split you actually used. Explain any preprocessing or augmentation that affects the interpretation. An annotated example can help if the task or label convention is unfamiliar.

3 Methods

Describe the baseline and your variant or variants. Identify the changes you made and **explain why you chose them**. For a pretrained model, identify the weights and the parts you trained.

4 Experimental Setup and Evaluation

Describe which parameters you actually trained, what varied across runs, and how you evaluated the results. **Include a training-loss curve from a named run and interpret it.** Keep the raw log and configuration with the code so the curve can be reproduced.

5 Results and Discussion

Present the main findings in the form that makes them clearest. Explain where the approach worked, where it struggled, or why a result surprised you. Distinguish an observed pattern from a possible explanation, particularly when several variables changed together.

6 Conclusion

Answer the question within the scope of your data and experiments. State what remains uncertain. If your findings led to a new idea, describe how you would investigate it.

References

Credit the datasets, methods, pretrained weights, and adapted code you used. Use a consistent citation style and provide enough detail for readers to find each source.

Ways to present experimental evidence

Beyond the **required training-loss curve**, choose the evidence that answers your question. The metrics, plots, and case studies on this page are suggestions. A short numerical comparison or a well-chosen visual can be enough to explain a finding.

Showing the task

A labeled image can explain what counts as a correct prediction. A box around a partly occluded pedestrian may clarify the detection task. An overlay of a pet mask can show how ambiguous boundary pixels are treated. If you use a visual, explain the colors or symbols a reader needs to interpret it.

Choosing a useful comparison

A table can show results across model versions, while a curve can show how performance changes across training-set sizes. Choose **your own measures and presentation**. If an improvement concerns a rare group, giving counts alongside percentages helps readers judge its scale.

Task	Possible metrics	Examples of supporting evidence
Classification	Accuracy, or macro-F1 if classes are imbalanced.	Per-class recall or a confusion matrix can show which classes the model struggles to distinguish.
Detection	AP50 for a single class, or mAP50 when averaging across classes. Both use a box-overlap threshold of 0.50.	Mark misses, false alarms, or poorly localized boxes. Use the same score threshold when comparing displayed predictions.
Segmentation	Foreground IoU or Dice for a binary mask, or mean IoU across named classes for a multiclass mask.	A contour overlay can reveal boundary errors that an average score hides.

Making an error interpretable

A missed sign in a dark scene can be more informative than a gallery of easy detections. You might compare predictions on that scene, then check whether similar failures recur on the held-out set. Select the examples and level of detail that help distinguish a pattern from an isolated case.

Before submission

Try the README instructions on your small test pack. Check that the named input produces the saved prediction and that the raw log regenerates the curve. **Explain what you personally changed, ran, and learned.**